Paleomagnetism of Steel Mountain and Indian Head anorthosites from western Newfoundland

G. S. MURTHY AND K. V. RAO

Geomagnetic Research Laboratory, Department of Physics, Memorial University of Newfoundland, St. John's, Newfoundland A1C 5S7

Received 9 April 1975

Revision accepted for publication 4 September 1975

As part of studies of anorthosites from the northern hemisphere, the paleomagnetism of two anorthosite inliers at Indian Head and Steel Mountain in western Newfoundland is described. The six sites at Indian Head retain a stable remanent magnetization and yield a paleomagnetic pole at 8° S, 157½° E ($\delta p = 15°$, $\delta m = 20°$). This is close to poles for anorthositic and other rock units from within the Grenville Province that represent magnetization of age 1000 m.y., suggesting that there is no relative rotation between western Newfoundland and the southeastern part of the Canadian Shield, at least since the time of the Grenville orogeny. The ten sites at Steel Mountain also retain a stable magnetization, which yields a paleomagnetic pole at 22½° S, 41° W ($\delta p = 8°$, $\delta m = 14°$). This position is closer to lower Paleozoic poles from Newfoundland and thus probably represents a lower Paleozoic magnetization.

C'est dans le cadre de l'étude des anorthosites de l'hémisphère septentrional qu'est considéré ici le paléomagnétisme des deux massifs anorthositiques d'Indian Head et Steel Mountain, dans l'ouest de Terre-Neuve.

Les six sites d'Indian Head montrent un magnétisme rémanent stable et indiquent un pôle paléomagnétique à 8° S, 157½° E ($\delta p = 15°$, $\delta m = 20°$). Ceci est proche des pôles des roches anorthositiques et autres de l'intérieur de la Province de Grenville, qui représentent un magnétisme datant de 1000 Ma; ceci suggère donc l'absence de rotation relative entre Terre-Neuve et la partie sud-est du bouclier canadien, au moins depuis l'époque de l'orogène de Grenville.

Les dix sites étudiés sur la Steel Mountain fournissent aussi un magnétisme stable indiquant un pôle paléomagnétique à 22½° S, 41° O ($\delta p = 8°$, $\delta m = 14°$). Cette position se rapproche davantage des pôles du Paléozoïque inférieur de Terre-Neuve et représente donc probablement un magnétisme de cet âge.

[Traduit par le journal]

Introduction

This investigation is part of a study of the anorthosite massifs of the northern hemisphere, including their magnetic properties. Herz (1969) has summarized most of the relevant material pointing to the uniqueness in time and space of these plutonic bodies. When plotted on maps of the presumed pre-drift paleogeography, most anorthosite massifs occur in two principal belts, one of which occupies landmasses in the present northern hemisphere, beginning in either the Ukraine or Finland and continuing westwards through Sweden, southern Norway, the Outer Hebrides, Greenland, and eastern Canada and the United States. The southern hemisphere anorthosite belt begins at Capivarito, Brazil, and extends east through Angola, Tanzania, Antarctica, the Eastern Ghats of India, and possibly into the Musgrave Ranges of Australia. A special feature of the anorthosite belts is that the ages of these rocks are all clustered around 1300 ± 200 m.y., except those within the Grenville Province of the Canadian Shield, which yield typical Grenville ages of around 1000 m.y. There is growing evidence that the anorthosites are the product of a unique world-wide event, perhaps occurring under thermal conditions in the upper layers of the earth that have not been duplicated since the Precambrian. A comprehensive study of the remanent magnetism of these bodies was started at the Memorial University of Newfoundland, with a view that comparison of the paleomagnetic poles of these approximately synchronous massifs from three landmasses bordering the North Atlantic (North America, Greenland, and Western Europe) would provide a powerful method for reconstructing the relative positions of the relevant landmasses. As part of this project, paleomagnetic studies are underway on the Egersund anortho-

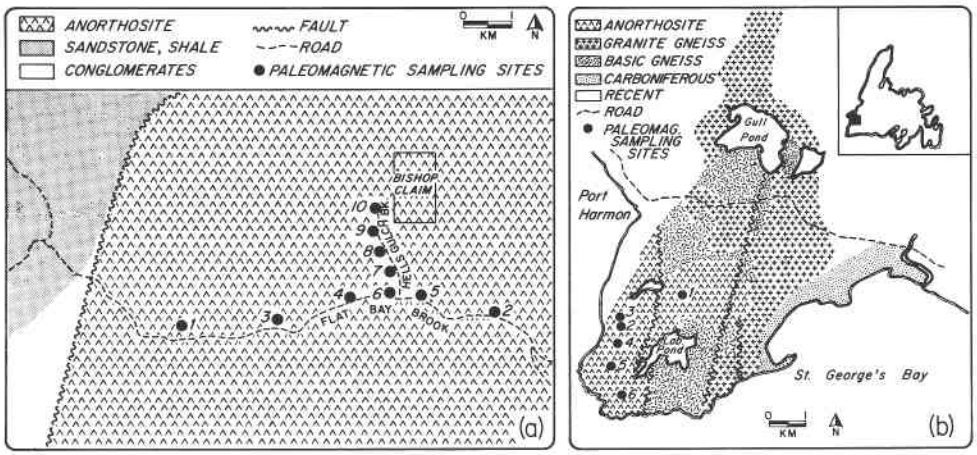


FIG. 1. Geology of the (a) Steel Mountain and (b) Indian Head areas (48½° N 58½° W) and the location of paleomagnetic sampling sites.

site of Norway (Murthy and Deutsch 1974) and on the Nain massif of Labrador.

Anorthosite Inliers in Newfoundland

Two anorthosite inliers, at Steel Mountain and at Indian Head in western Newfoundland, form part of Zone A or Lomond Zone of Williams *et al.* (1974). Both inliers, located at 48½° N latitude, 58½° W longitude, have undergone metamorphism to the granulite facies. They lie in the Precambrian basement rocks and are presumed to be equivalents of similar anorthosite bodies from within the Grenville Province.

The anorthosite body at Steel Mountain (Fig. 1a) is part of a complex of crystalline gneisses and intrusive rocks that form an elevated plateau, sloping gently eastward. The complex is in fault contact with Carboniferous sedimentary rocks on the west side. Within the complex, an elliptical body of anorthosite covering an area of approximately 600 sq. km lies east of the head of Bay St. Georges. Baird (1954) distinguishes four types of anorthosites in the area: (1) the Cairn Mountain type, containing a light to dark gray, coarse grained anorthosite; (2) the Eastern border facies type, a medium to fine grained anorthosite containing very pale pink feldspar; (3) gabbroic type anorthosite, very similar to most of the Labrador anorthosites; and finally (4) a pink and brown anorthosite that is intermediate in appearance between that of the Cairn Mountain type and the gabbroic type.

A smaller body of anorthosite occurs about 16 km to the northwest in the Indian Head peninsula (Fig. 1b), separated from the Steel Mountain body by sedimentary rocks of Carboniferous age. Heyl and Ronan (1954) and more recently Gross (1967), Rose (1969), and Colemann-Saad (1969) described the geology of the area. Heyl and Ronan suggest that the Indian Head is a disconnected western edge of the Long Range massif. The complex is in the form of a broad dome with shallow dipping limbs, that plunge 15° to 30° NE. The dome structure is disrupted by a number of faults that strike north to northeast. Iron and titanium oxide deposits are associated with the anorthosite.

A potassium–argon age determination is available for a chlorite extract from the gabbroic anorthosite of the Steel Mountain complex (Leech *et al.* 1963), which gave an age of 451 ± 35 m.y. Although Williams (1970) regarded the Steel Mountain complex as a Grenville inlier, he pointed out the obscurity of relationship of this anorthosite unit to the Paleozoic rocks. In his description of the regional geology around Stephenville, Riley (1962) ascribed a "pre-Carboniferous and probably post-Ordovician" age for the Long Range igneous and metamorphic complex in the area. It is possible that the chloritization is much younger than the host rock anorthosite and gives only a minimum age for the anorthosite.

The anorthosites of Indian Head peninsula seem to be much better dated. A biotite concentrate from the granite augen gneiss that is

part of the Indian Head intrusive complex gave a potassium–argon age of 830 ± 42 m.y. (Lowdon 1961). A sample of biotite from a pegmatite dike that cuts the interbanded titaniferous magnetite and gabbroic anorthosite at the Indian Head mine was dated at 900 ± 45 m.y. by potassium–argon methods (Lowdon *et al.* 1963). Thus, much better age control was available for this massif, and its similarity in age with other anorthosite massifs in the Grenville Province (St. Urbain, 890 ± 44 m.y.; Allard Lake, less than 1025 ± 60 m.y.; Magpie Mountain, 930 ± 46 m.y.) indicates that all these anorthosites underwent simultaneous metamorphism.

As part of the geologic investigations of titaniferous deposits of Canada, Rose (1969, 1972) reported some results of preliminary remanent magnetism measurements of titaniferous deposits associated with Indian Head and Steel Mountain anorthosites. His conclusions in part were that, in the Steel Mountain area, the remanent magnetism of titaniferous deposits is randomly oriented and subordinate to induced magnetism, reflecting the instability of magnetization of the deposits. In the Indian Head area, remanent magnetism is also not strongly retained in the titaniferous magnetite and gabbroic anorthosites.

Magnetic Measurements

Oriented samples were collected for the paleomagnetic study from anorthosites and gabbroic anorthosites. In the Steel Mountain area (Fig. 1a), most of the paleomagnetic sites were located along a traverse up Hells Gulch Brook from the north bank of Flat Bay Brook to Bishop Claim, where a presently abandoned titaniferous magnetite deposit occurs. Anorthosites become generally coarse grained toward Bishop Claim. Thirty-eight samples were collected at ten sites. In the Indian Head area (Fig. 1b), the anorthosites form the hilltop and are much less accessible. Twenty-one samples were collected at six sites. Paleomagnetic sampling sites are shown in Fig. 1.

From each oriented sample, at least two specimens, 2.5 cm in diameter and 2.4 cm in length, were obtained and used in the paleomagnetic measurements. Remanent magnetism measurements were made using a PAR spinner magnetometer. Measurement repeatability was usually to 1% in intensity and 1° in direction. AF demagnetization was carried out with a three-axis tumbler. The furnace for stepwise thermal demagnetization is non-inductive and residual fields were less than 50 γ. Curie points were determined using a vertical balance with an accuracy of ± 15 °C and were repeatable to ± 5 °C.

Steel Mountain

The NRM intensity values (J_n) varied between 12×10^{-4} and 5×10^{-7} emu/cm^3. The spread in direction was not as diverse, most of them being southeasterly with medium downward inclinations. Koenigsberger ratio values ($Q_n = J_n/kH$, where k is the susceptibility and H = 0.5 Oe) are typically around 1. Two types of behavior were noticeable during stepwise alternating field demagnetization (Fig. 2). The majority of specimens have behavior typified by S25 A, with median destructive fields around 50 peak oersteds. The response of another group of specimens was similar to S12 A, where the intensity drop even at maximum demagnetizing fields is relatively small. Cases are known in the literature (Irving 1964) where such an intensity decay is typical of remanence carried by hematite. The directional response (Fig. 2b) is such that either there is no change in direction with demagnetization, or best grouping was arrived at in fields of the order of 200 Oe. In either case, magnetization directions all clustered around southeasterly declinations and medium inclinations. Table 1 shows that the grouping, represented by Fisher's precision parameter K, has improved with AF demagnetization. One or two specimens from each site were subjected to thermal demagnetization (Fig. 2c,d) in a stepwise fashion. The change in direction in specimens was such that, after heating to successively higher temperatures, the directions either remained stable or moved progressively towards southeasterly declinations and medium inclinations. To test the possibility that a higher coercive force component, similar to that obtained by Irving *et al.* (1974*a*) for Morin anorthosite, is still present in these rocks, we have AF demagnetized two sites of Steel Mountain anorthosite at 900 Oe (the maximum alternating fields that can be achieved with our setup). The site mean directions are completely random at this stage, suggesting that there is no higher coercive force component present. Similar response to AF and thermal demagnetizations also

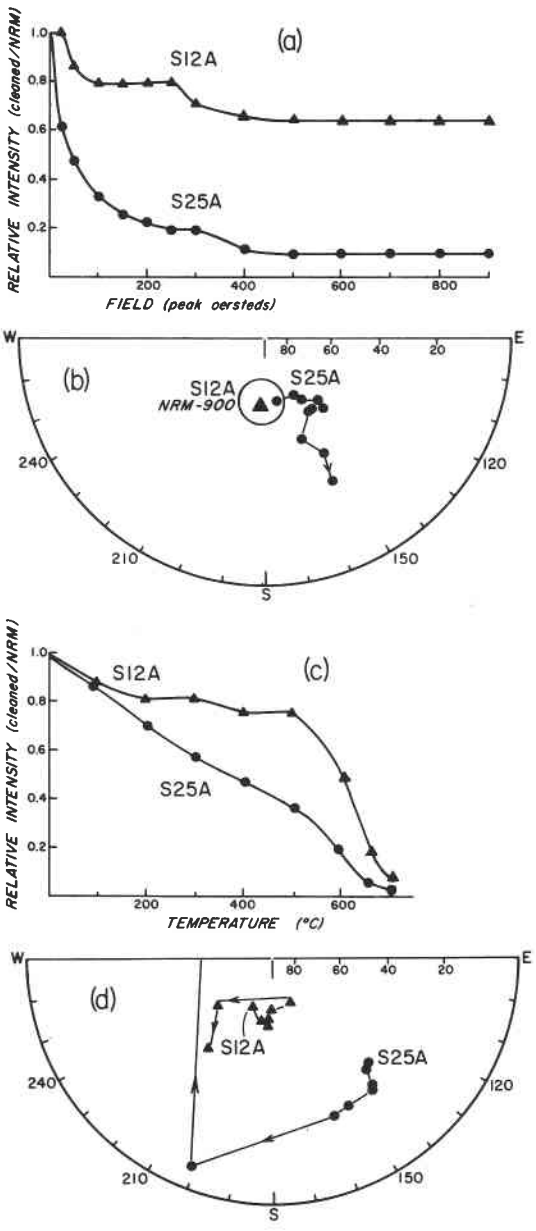


FIG. 2. Response of specimens from Steel Mountain anorthosite to alternating field (a, b) and thermal (c, d) demagnetizations. Wulff's net is used for plotting the changes in directions (b) and (d). Directions of S25 A are random beyond 600 Oe treatment (not shown). Solid symbols indicate positive inclinations.

suggests there was no high coercive force component not uncovered, and the directions obtained are those for the only component of magnetization that is retained by these rocks after optimum demagnetization at 200 Oe. The specimens are too weak magnetically to measure properly beyond the sixth step (600 °C) during thermal demagnetization (Figs. 2c and 2d), and this further confirms that there is no viable, high blocking temperature component still residing in these specimens. The directions of magnetization of site 2 are entirely anomalous. It is possible the samples of this site were from a locally displaced block. The results of this site were not included in the final calculations. Site 1 directions are close to those of the Indian Head anorthosite and are different from the rest of the sites of Steel Mountain. This site is the most westerly of all the sites and may be the only site at Steel Mountain carrying the original Grenvillian remanence direction. This is further discussed. Site mean directions of magnetization and the corresponding statistical parameters are presented in Table 1.

Indian Head

The NRM intensities of the Indian Head anorthosites were distributed between 6.0×10^{-2} and 2.0×10^{-6} emu/cm^3. Koenigsberger ratios are typically close to 1 within the range 0.5 to 20. The NRMs were analyzed by means of alternating field and thermal demagnetizations, and typical responses are shown in Fig. 3. Pro-

TABLE 1. Site mean directions of magnetization of Steel Mountain and Indian Head anorthosites

	NRM				After AF demagnetization*			
Site	N	D	I	K	D	I	K	α_{95}
				Steel Mountain				
1	5	309	−30½	<2	346½	−71	6	32
2	5	150	−½	10	143	−14½	12	23
3	5	166	61	7	142½	52	13	21
4	5	174	26½	7	177	15½	48	11
5	4	174	47½	14	161	30½	16	24
6	3	177½	25	19	165½	26½	16	32
7	3	165	33½	18	167	27½	47	18
8, 9	5	175½	31	6	178	32½	27	15
10	3	147½	43½	7	137	38½	44	19
				Indian Head				
2	3	78½	62½	11	335½	−56	28	24
1, 3	4	22	17	1	298	−63½	14	25
4	4	111	12	4	295½	−39½	24	19
5	4	340	−63	2	301	−65	40	15
6	6	325½	−44½	2	331	−62½	14	18

N = number of samples measured; D = declination of remanent magnetization; I = inclination of remanent magnetization; K = estimate of Fisher's k parameter; α_{95} = semi-angle of cone of 95% confidence for mean direction.

*Optimum fields of demagnetization are 200 peak oersteds for Steel Mountain and 300 peak oersteds for Indian Head samples.

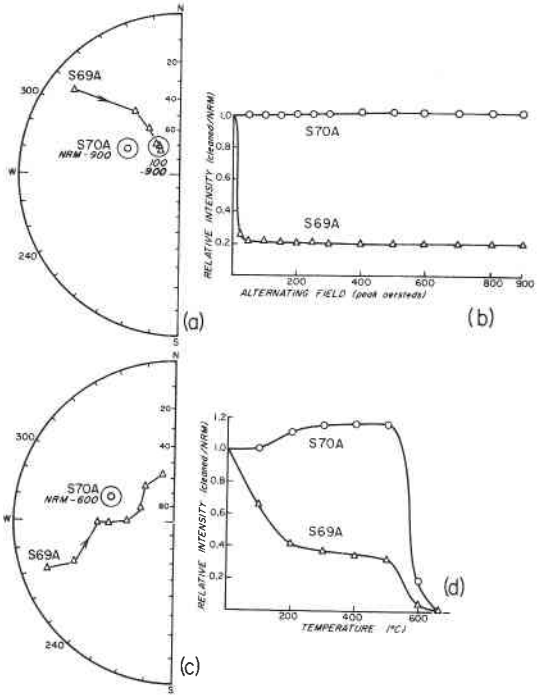


FIG. 3. Response of specimens from Indian Head anorthosite to alternating field (a, b) and thermal (c, d) demagnetizations. Wulff's net is used for plotting the directional changes in (a) and (c). Open symbols indicate negative inclinations.

gressive demagnetization of several pilot specimens of each site showed that the magnetization gradually changed in direction in alternating fields up to 300 Oe, owing to the elimination of viscous components. Upon further treatment at higher alternating fields, the direction of the decreasing NRM remained steady, proving that only one component was left after elimination of the viscous part of remanence. Particularly notable was the change in direction for some specimens, from positive inclinations to steep negative inclinations; this was seen both in AF and thermal demagnetizations. The stability of the magnetization of the rocks during the demagnetization procedures may indicate that the observed magnetization is most probably acquired when the rocks have undergone granulite facies metamorphism during Grenville orogeny. It should be noted, however, that during thermal demagnetization of some specimens (typified by S69 A in Fig. 3), no stable end point was attained. The rest of the specimens in this collection were AF demagnetized at 300 Oe, resulting in a much better grouping at all levels, and particularly the site mean directions of magnetization. Table 1 gives the results before and after AF cleaning and the corresponding statistical parameters. All samples from three sites of Indian Head anorthosite were AF demagnetized at 900 Oe to see if any improvement in grouping of site means could be achieved. The directions became randomly grouped after this demagnetization, indicating that the optimum field demagnetization at 300 Oe is best for this rock unit.

Magnetic Mineralogy

Polished sections of samples both from Indian Head and Steel Mountain were prepared and studied using a Zeiss optical microscope at $\times 1000$ magnification. The opaques, which are quite scarce, are usually elongated in shape (typical sizes $1 \times 5\ \mu$m, as shown in Fig. 4), similar to those observed in other anorthosite bodies (Murthy *et al.* 1971; Palmer and Carmichael 1973; Hargraves and Fish 1972; and Irving *et al.* 1974*a*). Most of the samples have, in addition, very fine grains distributed in a nonmagnetic matrix. Typical sizes of these small grains are of the order of a fraction of a micrometre. We were not able to ascertain from microscopic studies whether these small grains and slightly larger rods are magnetite or hematite, because of their small sizes. While Murthy *et al.* (1971) inferred from their magnetic experiments that the rod-shaped grains in anorthosites are probably magnetite, Hargraves and Fish (1972) and Irving *et al.* (1974*a*) had identified them as being hematite or ilmenohematite. In addition to these fine grains, S69 also contains large discrete grains of magnetite (about 100 μm).

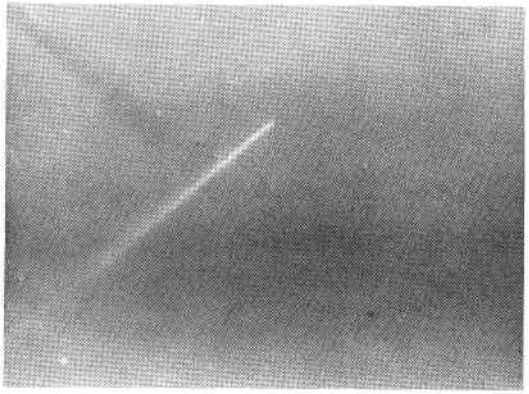


FIG. 4. Photomicrograph showing typical needle-shaped opaque grain in S70. The photomicrograph is 0.06 mm long.

The response of this sample to AF demagnetization, where an initial drop to 20% in original intensity is followed by a steady ratio of J/J_0 (Fig. 3b), is consistent with the presence of these large grains of magnetite. However, the presence of large grains is confined only to this particular sample. Curie point measurements were made of one sample each from the two localities, using a Curie balance in fields of the order of 1300 Oe. Typical curves are shown in Fig. 5. Curie points (defined according to convention) are 650 °C from the heating curves and around 610 °C from the cooling curves. Both lower Curie points and lower magnetic moments were observed after heating. This is probably due to chemical changes in the powdered samples, although it is possible that the lower Curie temperatures may be the result of temperature lag in the experiments. Blocking temperatures from thermal demagnetization experiments (Figs. 2c and 3d) are in the range 600–650 °C. These are slightly lower than if hematite were the remanence carrier, but are comparable to values obtained by Hargraves and Fish (1972) and Irving *et al.* (1974*b*). We have also studied the saturation isothermal remanence of these samples to infer the nature of the magnetic mineral. Roquet (1954) showed that isothermal remanence of specimens containing magnetite saturate in fields of the order of 3000 Oe, while hematite-containing specimens require fields of about 30 000 Oe for saturation. This provides a simple test that distinguishes hematite- and magnetite-containing specimens, although it could fail to detect small quantities of hematite in the presence of magnetite. All the specimens studied this way, with one exception, had their isothermal remanences still increasing beyond application of 8000 Oe, suggesting hematite as the most probable carrier of remanence. The exception is S69, which more or less reached saturation in fields of 4000 Oe.

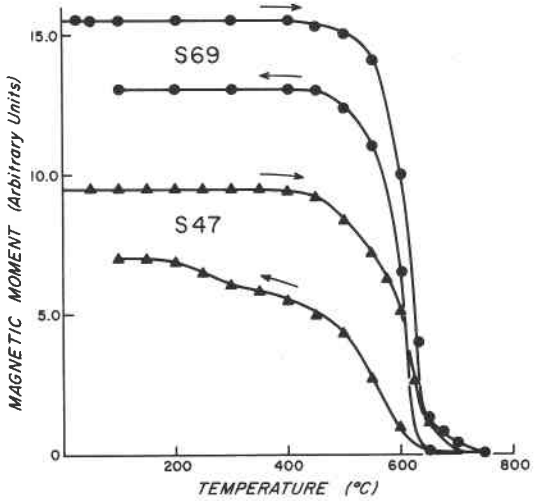


FIG. 5. Curie point determinations of powdered samples from Steel Mountain (S47) and Indian Head (S69) areas.

Discussion of Results and Conclusions

The site mean directions after alternating field demagnetization were combined to give mean directions of magnetizations for each of the two intrusions. The parameters of these mean directions are:

Steel Mountain: N = 7 sites, D = 163,° I = +32½°, K = 22 and α_{95} = 13°

Indian Head: N = 5 sites, D = 311°, I = −58½°, K = 30 and α_{95} = 14°

where the symbols are the same as in Table 1. The mean directions for the two localities are plotted in Fig. 6. The angular dispersion $\Delta = 81°/K^{\frac{1}{2}}$ is 17° and 14.7°, respectively, for the two rock units. The use of confidence limits for the precision parameter K (Cox 1969) provides upper limits for values of Δ of 25.4° and 23.2°, respectively. These values are greater than those calculated by Creer (1962) for the present earth's field at low latitudes, indicating that the sampling has been sufficiently detailed in the two localities that the paleosecular variation is fully represented in the measurements. The paleomagnetic poles corresponding to these mean directions are 22½° S, 41° W ($\delta p = 8°$, $\delta m = 14°$) for the Steel Mountain complex and at 8° S, 157½° E ($\delta p = 15°$, $\delta m = 20°$) for the Indian Head anorthosite. The normal pole for Indian Head and the reversed pole for Steel Mountain (22½° N, 139° E) are plotted in Fig. 7, along with poles for anorthosites and other rock units from within the Grenville Province, which represent magnetization around 1000 m.y.

Except for poles 6A, 8C, and 9B, the rest of the poles in Fig. 7 seem to fall in two clusters, one at the southern tip of Australia and a more predominant one around the equatorial latitudes at 160° east longitude. Irving *et al.* (1974*b*),

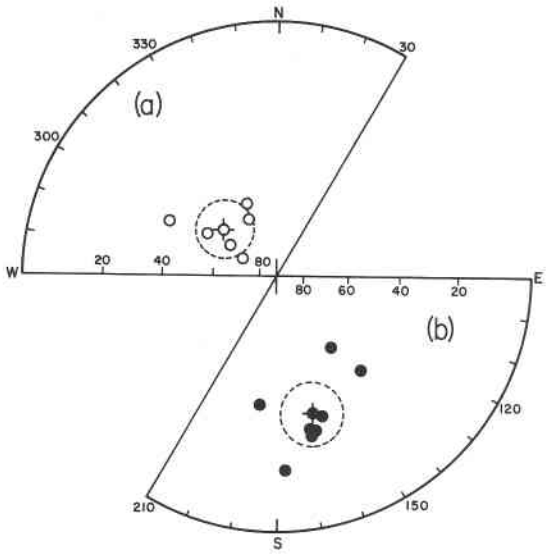


FIG. 6. Site mean directions after optimum AF cleaning for (a) Steel Mountain and (b) Indian Head anorthosites plotted on a Wulff net. Dashed circles are the α_{95} circles of confidence around the mean direction. Solid symbols indicate positive inclinations and open symbols indicate negative inclinations.

while discussing the magnetization of the Morin complex (poles 2A and 2B of Fig. 7), drew a Grenville polar track from one group of poles to the other and estimated the age span of the track as approximately 1124 (2A) to 1000 (2B) m.y. They also suggest reasons why poles 6A, 8C, and 9B do not fall on this track. While it seems from Fig. 7 that it is desirable to have some intermediate points on the polar track between the two clusters, there seems to be no doubt about the main grouping of the Grenville poles representing 1000 m.y. magnetization around equatorial latitudes at 160° east longitude. The close similarity of the Indian Head pole with this main grouping of Grenville poles is striking. This similarity indicates that there was no relative movement between the southeastern part of the Canadian Shield (the Grenville Province) and western Newfoundland, at least from the time the measured magnetization of the Indian Head anorthosites was imprinted; *i.e.*, during the times of the Grenville orogeny. Black (1964) postulated that comparisons of paleomagnetism of Cambrian and later formations from Newfoundland and New Brunswick suggest that Newfoundland had rotated anticlockwise by 30° during mid-Devonian times, relative to mainland North America. This was later questioned by Robertson *et al.* (1968), Pearce and Freda (1969), and Deutsch (1969). The Indian Head pole was rotated about the northern pivot suggested by Black, and the new position of the pole was at 1° S, 135° E, significantly shifted away from the Grenville poles. This result suggests that it is improbable that relative rotation took place between western Newfoundland and the Grenville Province part of the Canadian Shield, at least since the time of the Grenville orogeny.

The Steel Mountain pole is significantly farther away from the Grenville poles. This pole falls on

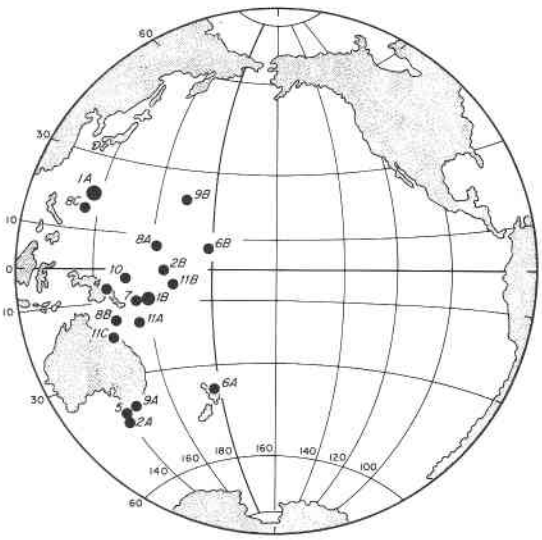


FIG. 7. Paleomagnetic pole positions for Steel Mountain and Indian Head anorthosites, plotted along with poles for anorthositic and other rock units from within the Grenville Province of the Canadian Shield representing magnetization of around 1000 m.y. 1A, Steel Mountain (this paper); 1B, Indian Head (this paper); 2A, Morin complex anorthosite, hcf component (Irving *et al.* 1974*a*); 2B, Morin complex anorthosite, mcf component (Irving *et al.* 1974*a*); 4, St. Urbain anorthosite (Hargraves and Roy 1974); 5, Allard Lake anorthosite (Hargraves and Burt 1967); 6A, Mealy Mountain anorthosite, east component (Fahrig *et al.* 1974); 6B, Mealy Mountain anorthosite, northwesterly component (Fahrig *et al.* 1974); 7, Lac St. Jean anorthosite (Fahrig *et al.* 1974); 8A, Grenville Front anorthosite (Palmer and Carmichael 1973); 8B, Wilberforce pyroxenite (Palmer and Carmichael 1973); 8C, Tudor Gabbro (Palmer and Carmichael 1973); 9A, Haliburton basic rocks, group A (Buchan and Dunlop 1973); 9B, Haliburton basic rocks, group B (Buchan and Dunlop 1973); 10, Grenville dikes (Murthy 1971); 11A, Whitestone anorthosite, hematite component (Ueno *et al.* 1975); 11B, Whitestone anorthosite, magnetite component (Ueno *et al.* 1975): 11C, Whitestone diorite (Ueno *et al.* 1975).

the extension of the polar track relative to Laurentia after suture of Grenvillia with interior Laurentia as suggested by Irving *et al.* (1974*b*), indicating that the magnetization of the rock unit could possibly be younger than 700 m.y. It is probable that the Steel Mountain pole corresponds to the 451 m.y. date, and that the magnetization that the Steel Mountain anorthosite possesses is only a later secondary magnetization, corresponding to the 451 m.y. event that the potassium–argon age dating records. The remanent intensities of Steel Mountain anorthosites are extremely weak, possibly suggesting the alteration of the magnetic mineral. Further, there is better agreement for this pole position with lower Paleozoic poles from Newfoundland (Beals *et al.* 1974; E. R. Deutsch, personal communication 1975) than with the Grenville poles.

The virtual geomagnetic pole for site 1 of Steel Mountain was at 14½° N, 130½° E, much closer to the Grenville poles than the paleomagnetic pole derived from the other seven sites. Site 1 is the most westerly of all sites in this area and seems to have undergone the least alteration. This indicates possibly that only at this site is the Grenville magnetization retained in the Steel Mountain anorthosite.

It is also of interest to note that the remanence direction of the Indian Head anorthosite is similar to that of farsundite (Murthy and Deutsch 1975) and of Egersund anorthosite (Murthy and Deutsch 1974) from south Rogaland in Norway, after correction for closure of the North Atlantic and of other anorthosite massifs from within the Grenville Province. This suggests that all these massifs had undergone similar metamorphism during the Grenville orogeny.

Acknowledgments

The authors are grateful to Dr. E. R. Deutsch for critically reading the manuscript and offering suggestions for improvement. Mr. R. Neale helped in the field collection. This work has been supported by the Department of Energy, Mines and Resources Research Agreement and by the National Research Council of Canada Operating Grant A8707 to G.S.M.

Baird, D. M. 1954. Magnetite and gypsum deposits of the Sheep Brook – Lookout Brook area. Geol. Surv. Can. Bull. 27, pp. 20–41.

Beals, F. W., Carracedo, J. C., and Strangway, D. W. 1974. Paleomagnetism and the origin of Mississippi Valley-type ore deposits. Can. J. Earth Sci. **11**, p. 211–223.

Black, R. F. 1964. Paleomagnetic support of the theory of rotation of the western part of the island of Newfoundland. Nature, **202**, pp. 945–948.

Buchan, K. L. and Dunlop, D. J. 1973. Magnetization episodes and tectonics of the Grenville Province. Nature Phys. Soc. **246**, pp. 28–30.

Coleman-Sadd, S. P. 1969. Geology of the iron deposits near Stephenville, Newfoundland. MSc Thesis, Memorial Univ. Newfoundland, St. John's, Newfoundland. 97 p.

Cox, A. 1969. Confidence limits for the precision parameter K. Geophys. J. R. Astron. Soc. **18**, pp. 545–549.

Creer, K. M. 1962. The dispersion of the geomagnetic field due to secular variation and its determination for remote times from palaeomagnetic data. J. Geophys. Res. **67**, pp. 3461–3476.

Deutsch, E. R. 1969. Paleomagnetism and North Atlantic Paleogeography. *In*: North Atlantic—geology and continental drift. (M. Kay, *Ed.*) AAPG Mem. 12, pp. 931–954.

Fahrig, W. F., Christie, K. W., and Schwarz, E. J. 1974. Paleomagnetism of the Mealy Mountain anorthosite suite and of Shabogamo gabbro, Labrador, Canada. Can. J. Earth Sci. **11**, p. 18–29.

Gross, G. A. 1967. Geology of iron deposits in Canada, Vol. 2, Iron deposits in the Appalachian and Grenville regions of Canada. Geol. Surv. Can. Econ. Geol. Rep. 22, 111 p.

Hargraves, R. B. and D. M. Burt 1967. Paleomagnetism of the Allard Lake anorthosite suite. Can. J. Earth Sci. **4**, pp. 357–369.

Hargraves, R. B. and Fish, J. R. 1972. Paleomagnetism of anorthosite in southern Norway and comparison with an equivalent in Quebec. Geol. Soc. Am. Mem. 132, pp. 141–148.

Hargraves, R. B. and Roy, D. W. 1974. Paleomagnetism of anorthosite in and around the Charlevoix cryptoexplosion structure, Quebec. Can. J. Earth Sci. **11**, pp. 854–857.

Herz, N. 1969. Anorthosite belts, continental drift, and the anorthosite event. Science, **164**, pp. 944–947.

Heyl, A. V. and Ronan, J. J. 1954. The iron deposits of Indian Head area. Geol. Surv. Can. Bull. 27, pp. 42–65.

Irving, E. 1964. Paleomagnetism and its application to geological and geophysical problems. John Wiley & Sons, New York. 399 p.

Irving, E., Park, J. K., and Emslie, R. F. 1974*a*. Paleomagnetism of the Morin complex. J. Geophys. Res. **79**, pp. 5482–5490.

Irving, E., Emslie, R. F., and Ueno, H. 1974*b*. Upper Proterozoic paleomagnetic poles from Laurentia and the history of the Grenville structural Province. J. Geophys. Res. **79**, pp. 5491–5502.

Leech, G. B., Lowdon, J. A., Stockwell, C. H., and Wanless, R. K. 1963. Age determinations and geological studies. Geol. Surv. Can. Pap. 63-17, 140 p.

Lowdon, J. A. 1961. Age determinations by the Geological Survey of Canada. Geol. Surv. Can. Pap. 61-17, 127 p.

Lowdon, J. A., Stockwell, C. H., Tipper, H. W., and Wanless, R. K. 1963. Age determinations and geological studies. Geol. Surv. Can. Pap. 62-17, 140 p.

Murthy, G. S. 1971. The paleomagnetism of diabase

dikes from the Grenville Province. Can. J. Earth Sci. **8**, pp. 802–812.

MURTHY, G. S. and DEUTSCH, E. R. 1974. Rock magnetic and paleomagnetic investigations of anorthositic rocks from southern Norway. EOS, Trans. AGU, **55**, p. 234.

——— 1975. A new Precambrian paleomagnetic pole for northern Europe. Phys. Earth Planet. Int. (in press).

MURTHY, G. S., EVANS, M. E., and GOUGH, D. I. 1971. Evidence of single-domain magnetite in the Michikamau anorthosite. Can. J. Earth Sci. **8**, pp. 361–370.

PALMER, H. and CARMICHAEL, C. M. 1973. Paleomagnetism of some Grenville Province rocks. Can. J. Earth Sci. **10**, pp. 1175–1190.

PEARCE, G. W. and FREDA, G. N. 1969. Magnetization of the Perry formation of New Brunswick, and the rotation of Newfoundland: Discussion. Can. J. Earth Sci. **6**, pp. 353–355.

RILEY, G. C. 1962. Stephenville map-area, Newfoundland. Geol. Surv. Can. Mem. 323, 72 p.

ROBERTSON, W. A., ROY, J. L., and PARK, J. K. 1968. Magnetization of the Perry formation of New Brunswick, and the rotation of Newfoundland. Can. J. Earth Sci. **5**, pp. 1175–1181.

ROQUET, J. 1954. Sur les remanences des oxydes de fer et leur interet en geomagnetisme. Ann. Geophys. **10**, pp. 226–247.

ROSE, E. R. 1969. Geology of titanium and titaniferous deposits of Canada. Geol. Surv. Can. Econ. Geol. Rep. 25, 177 p.

ROSE, E. R. 1972. Notes on remanent magnetism associated with some titaniferous deposits in Canada. Proc. Geol. Assoc. Can. **24**, pp. 69–74.

UENO, H., IRVING, E., and MCNUTT, R. H. 1975. Paleomagnetism of the Whitestone anorthosite and diorite, the Grenville polar track, and relative motions of the Laurentian and Baltic shields. Can. J. Earth Sci. **12**, pp. 209–226.

WILLIAMS, H. 1970. Geology of southeastern Canada. *In*: Geology and economic minerals of Canada. (R. J. W. Douglas, *Ed*.) Econ. Geol. Rep. 1, p. 231.

WILLIAMS, H., KENNEDY, M. J., and NEALE, E. R. W. 1974. The northeastward termination of the Appalachian orogen. *In*: The ocean basins and margins, Vol. 2. (A. E. M. Nairn and F. G. Stehli, *Eds*.) Plenum Press, New York, pp. 79–123.